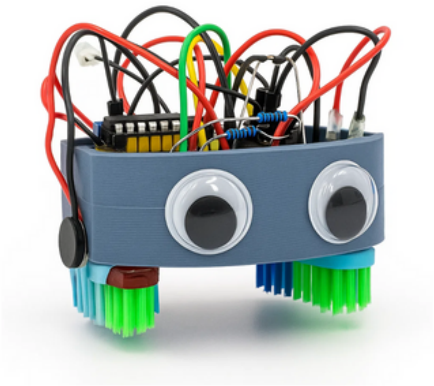
INTELLIGENT BRISTLE-BOT SWARM RECONNAISSANCE

01. The Smart Bug

The Smart Bug is a compact, low-cost swarm robotics kit built for educational purposes. Integrated with BLE localisation, magnetometer orientation, and simple vibration motor locomotion, it is capable of reconaissance and object detection within an indoor area. Acting as field agents, the collective objective of the swarm can be easily modified through software and hardware, easily accessible on the bots breadboard (i.e., brain). A web interface has been created to keep track of the beacon and agent positions during operation.

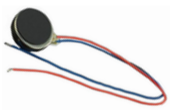


02. Key Specifications

- **Seeed Studio XIAO nRF52840 Sense**
  - Communication: BLE 5.0
  - Audio Detection: PDM Microphone
  - Motion Tracking: IMU

- **QMC5883L Magnetometer**
  - Orientation recognition

- **3.0V Vibration Motor**
  - Excites the bristles



- **2.7V LiPo battery**
  - Power the system

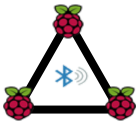
- **Toothbrush Heads**
  - The legs

03. Locomotion

- Two mini vibration motor controlled via GPIO pins
- Differential steering for navigation and steering
- Speed ~0.1 m/s

04. BLE Localisation

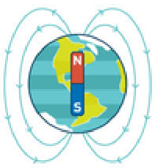
- Weighted trilateration using the RSSI values of 3 BLE beacons and free-space pathloss model
- Typical accuracy: ±0.25m in a 1.5m x 1m area



$$PL(d) - PL(d_o) = 10n\log(d/d_o)$$

05. Orientation

- Magnetometer: finds magnetic north
- Calibration process: Required
- Maximum deviation: ±10°



06. System Housing

- Net Weight:
  - ~62 g
- Limiting Dimensions:
  - Length 70 mm,
  - Width 60 mm,
  - Height 21 mm.
- 3D-printed PLA casing
  - Biodegradable plastic